

Which of these sentences are propositions? What are the truth values of those that are propositions?

Boston is the capital of Massachusetts.

Miami is the capital of Florida.

$2 + 3 = 5$.

$5 + 7 = 10$.

$x + 2 = 11$.

Answer this question.

Do not pass go.

What time is it?

There are no black flies in Maine.

$4 + x = 5$.

The moon is made of green cheese.

$2n \geq 100$.

What is the negation of each of these propositions?

Mei has an MP3 player.

There is no pollution in New Jersey.

$2 + 1 = 3$.

The summer in Maine is hot and sunny.

Jennifer and Teja are friends.

There are 13 items in a baker's dozen.

Abby sent more than 100 text messages every day.

121 is a perfect square.

Suppose that during the most recent fiscal year, the annual revenue of Acme Computer was 138 billion dollars and its net profit was 8 billion dollars, the annual revenue of Nadir Software was 87 billion dollars and its net profit was 5 billion dollars, and the annual revenue of Quixote Media was 111 billion dollars and its net profit was 13 billion dollars. Determine the truth value of each of these propositions for the most recent fiscal year.

Quixote Media had the largest annual revenue.

Nadir Software had the lowest net profit and Acme Computer had the largest annual revenue.

Acme Computer had the largest net profit or Quixote Media had the largest net profit.

If Quixote Media had the smallest net profit, then Acme Computer had the largest annual revenue.

Nadir Software had the smallest net profit if and only if Acme Computer had the largest annual revenue.

Let p and q be the propositions “Swimming at the New Jersey shore is allowed” and “Sharks have been spotted near the shore,” respectively. Express each of these compound propositions as an English sentence.

a) $\neg q$

b) $p \wedge q$

c) $\neg p \vee q$

d) $p \rightarrow \neg q$

e) $\neg q \rightarrow p$

f) $\neg p \rightarrow \neg q$

g) $p \leftrightarrow \neg q$

h) $\neg p \wedge (p \vee \neg q)$

Let p and q be the propositions

p : It is below freezing.

q : It is snowing.

Write these propositions using p and q and logical connectives (including negations).

a) It is below freezing and snowing.

b) It is below freezing but not snowing.

c) It is not below freezing and it is not snowing.

d) It is either snowing or below freezing (or both).

e) If it is below freezing, it is also snowing.

f) Either it is below freezing or it is snowing, but it is not snowing if it is below freezing.

g) That it is below freezing is necessary and sufficient for it to be snowing.

Let p , q , and r be the propositions

p :You get an A on the final exam.

q :You do every exercise in this book.

r :You get an A in this class.

Write these propositions using p , q , and r and logical connectives (including negations).

a) You get an A in this class, but you do not do every exercise in this book.

b) You get an A on the final, you do every exercise in this book, and you get an A in this class.

c) To get an A in this class, it is necessary for you to get an A on the final.

d) You get an A on the final, but you don't do every exercise in this book; nevertheless, you get an A in this class.

e) Getting an A on the final and doing every exercise in this book is sufficient for getting an A in this class.

f) You will get an A in this class if and only if you either do every exercise in this book or you get an A on the final.

State the converse, contrapositive, and inverse of each of these conditional statements.

If it snows today, I will ski tomorrow.

Converse:

Contrapositive:

Inverse:

I come to class whenever there is going to be a quiz.

Converse:

Contrapositive:

Inverse:

A positive integer is a prime only if it has no divisors other than 1 and itself.

Converse:

Contrapositive:

Inverse:

If it snows tonight, then I will stay at home.

Converse:

Contrapositive:

Inverse:

I go to the beach whenever it is a sunny summer day.

Converse:

Contrapositive:

Inverse:

When I stay up late, it is necessary that I sleep until noon.

Converse:

Contrapositive:

Inverse:

Translate the given statement into propositional logic using the propositions provided.

1. You cannot edit a protected Wikipedia entry unless you are an administrator.

Express your answer in terms of e : "You can edit a protected Wikipedia entry" and a : "You are an administrator."

2. You can see the movie only if you are over 18 years old or you have the permission of a parent.

Express your answer in terms of m : "You can see the movie," e : "You are over 18 years old," and p : "You have the permission of a parent."

3. You can graduate only if you have completed the requirements of your major and you do not owe money to the university and you do not have an overdue library book.

Express your answer in terms of g : "You can graduate," m : "You owe money to the university," r : "You have completed the requirements of your major," and b : "You have an overdue library book."

4. To use the wireless network in the airport you must pay the daily fee unless you are a subscriber to the service.

Express your answer in terms of w : "You can use the wireless network in the airport," d : "You pay the daily fee," and s : "You are a subscriber to the service."

5. You are eligible to be President of the U.S.A. only if you are at least 35 years old, were born in the U.S.A, or at the time of your birth both of your parents were citizens, and you have lived at least 14 years in the country.

Express your answer in terms of e : "You are eligible to be President of the U.S.A.," a : "You are at least 35 years old," b : "You were born in the U.S.A.," p : "At the time of your birth, both of your parents were citizens," and r : "You have lived at least 14 years in the U.S.A."

6. You can upgrade your operating system only if you have a 32-bit processor running at 1 GHz or faster, at least 1 GB RAM, and 16 GB free hard disk space, or a 64-bit processor running at 2 GHz or faster, at least 2 GB RAM, and at least 32 GB free hard disk space.

Express your answer in terms of u : "You can upgrade your operating system," b_{32} : "You have a 32-bit processor," b_{64} : "You have a 64-bit processor," g_1 : "Your processor runs at 1 GHz or faster," g_2 : "Your

processor runs at 2 GHz or faster,” $r1$: “Your processor has at least 1 GB RAM,” $r2$: “Your processor has at least 2 GB RAM,” $h16$: “You have at least 16 GB free hard disk space,” and $h32$: “You have at least 32 GB free hard disk space.”

Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent by developing a series of logical equivalences.

Show that each of these conditional statements is a tautology without using truth tables.

a) $(p \wedge q) \rightarrow p$

b) $p \rightarrow (p \vee q)$

c) $\neg p \rightarrow (p \rightarrow q)$

d) $(p \wedge q) \rightarrow (p \rightarrow q)$

e) $\neg(p \rightarrow q) \rightarrow p$

$$f) \neg(p \rightarrow q) \rightarrow \neg q$$

$$g) (p \wedge q) \rightarrow (p \vee q)$$

Express these system specifications using the propositions p “The user enters a valid password”, q “Access is granted”, and r “The user has paid the subscription fee” and logical connectives (including negations).

“The user has paid the subscription fee, but does not enter a valid password.”

“Access is granted whenever the user has paid the subscription fee and enters a valid password.”

“Access is denied if the user has not paid the subscription fee.”

“If the user has not entered a valid password but has paid the subscription fee, then access is granted.”

For each of these sentences, determine whether an inclusive or, or an exclusive or, is intended. Explain your answer.

a) Coffee or tea comes with dinner.

- b)** A password must have at least three digits or be at least eight characters long.
- c)** The prerequisite for the course is a course in number theory or a course in cryptography.
- d)** You can pay using U.S. dollars or euros.
- e)** To take discrete mathematics, you must have taken calculus or a course in computer science.
- f)** When you buy a newcar fromAcme Motor Company, you get \$2000 back in cash or a 2% car loan.
- g)** Dinner for two includes two items from column A or three items from column B.
- h)** School is closed if more than 2 feet of snow falls or if the wind chill is below -100 .

Write each of these statements in the form “if p , then q ” in English.

- a)** It snows whenever the wind blows from the northeast.
- b)** The apple trees will bloom if it stays warm for a week.
- c)** That the Pistons win the championship implies that they beat the Lakers.
- d)** It is necessary to walk 8 miles to get to the top of Long’s Peak.
- e)** To get tenure as a professor, it is sufficient to beworldfamous.

- f)** If you drive more than 400 miles, you will need to buy gasoline.

- g)** Your guarantee is good only if you bought your CD player less than 90 days ago.

- h)** Jan will go swimming unless the water is too cold.